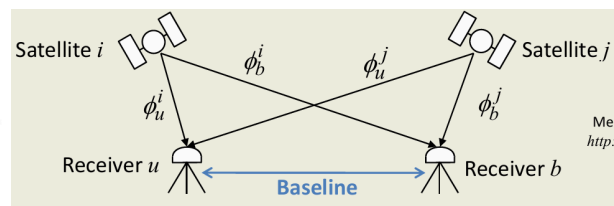
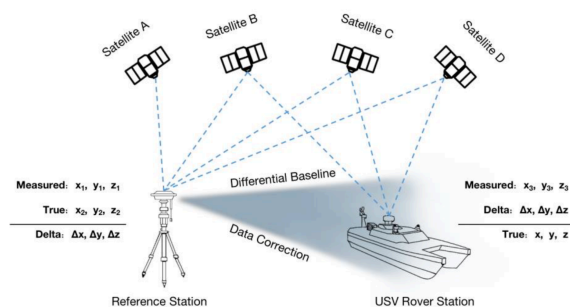


PRECISE HEIGHTS: DGNSS

Workflow Documentation

In gravimetric surveys, vertical accuracy is the primary constraint for data reduction. The Earth's gravity field varies with altitude according to the Free-Air Gradient (~ -0.3086 mGal/m). To ensure the reliability of the resulting Bouguer anomalies, an elevation accuracy of a few centimeters is required to keep the associated uncertainty below 0.01 mGal, which corresponds to the sensitivity of modern gravimeters. Traditional GPS elevations are insufficient, necessitating differential processing.

The survey employed the principle of **Carrier-Phase Differential GNSS**. This technique relies on a fixed **Base Station** with precisely known absolute coordinates and a mobile **Rover** receiver at the stations. By recording satellite signals simultaneously at both locations, "common-mode errors" (such as satellite clock biases and atmospheric delays) are mathematically cancelled out through **Double Differencing**. This allows the relative vector (baseline) between the Base and the Rover to be determined with centimeter-level precision.



Principle of Differential GNSS

Figures from : [A LiDAR SLAM-Assisted Fusion Positioning Method for USVs](#) & [Microsoft PowerPoint - GPS RTKLIB Seminar 2.ppt \[互換モード\]](#)

The processing was conducted using the last version of the open-source software **RTKLIB** 2.5.0 (Takasu, 2013). The heights obtained refer to Ellipsoidal Height (WGS84).

1. RTKLIB

Download the software here : [Release RTKLIB-EX 2.5.0 · rtklibexplorer/RTKLIB · GitHub](#)

Open the program : rtkpost.exe

1.1 Downloading the input files

The input files required are : rover observations, base station observation file, base station navigation file.

The input files optional are : auxiliary files.

They all have to be on the same day. [GPS Calendar - National Geodetic Survey](#)

Rover observations

Field data from the GPS receiver. This contains the raw distance measurements between the antenna and the satellites. Format in .o, .d, .rnx.

The raw field data was converted from .tps format to .o format (with TPS2RNX) to be used in RTKLIB.

For this survey the rover observations used are in the file */ 01_DGNSS_positionning / 1_Input_files / 1_Data_Rover_observation*.

Base station files (observation and navigation : MO & MN)

- Base station precise coordinates :

A base station is a permanent GPS station with fixed, perfectly known **coordinates** and heights. Stations **IZAN**, **TN01**, and **TN02** were selected as bases based on their proximity to the survey areas to minimize baseline length and atmospheric decorrelation.

Because the IGN coordinates were from 2008, more recent coordinates were taken from the NGL (Nevada Geodetic Library) [MAGNET + Global GPS Network Map](#) ; find the base station - click on tenv3 for the time table. Select the coordinates during the survey : 7 april to 15 oct 2024 and make a mean to obtain one mean value of the coordinate of the base station during the 2024 survey.

In RTKLIB options/position, enter manually the coordinates of the base station used.
/01_DGNSS_positionning/1_Input_files/2_Base_stations/base_station_NGL_2024_coordinates.csv

Station	Latitude (°)	Longitude (°)	Elevation ellispoidal (m)	Antenna height
TN01	28,47718693	-16,24115170	51,75309	0.0771
TN02	28,41825745	-16,55078448	54,41445	0.0771
IZAN	28,30806941	-16,49967718	2417,40444	0

- Base station files :

IGN daily Base Station files (from ERGNSS network) : [Index of /ERGNSS/diario_30s](#); find the date of the survey - download the Mixed Observations (MO.crx) and Merged Navigation (MN.rnx) files of selected base station for that specific survey day.

Observation (MO.crx) files were decompressed into standard RINEX (.rnx), using the converter (download it) : [CRX2RNX](#) (Hatanaka, 2024).

The MO and MN files used for each base stations are in .rnx in :
/ 01_DGNSS_positionning / 1_Input_files / 2_Base_stations / IZAN or TN01 or TN02

Auxiliary / Corrections files : SP3, CLK, ERP & ATX

For better precision and corrections, download the auxiliary files from NASA (create account) : [CDDIS](#) | [archive](#) | [gnss](#) ; select the GPS week of your survey - search for **IGS00PSFIN** (International GNSS Service, world reference & final official data) - select the survey date - download the following files : **SP3** (precise ephemeris), **CLK** (precise clock) and **ERP** (Earth rotation corrections) which is 1 per week to will be at the first day of the week.

Also, antenna phase center corrections need to be applied : download the *igs20.atx* file from [Index of /pub/station/general](#), which is the absolute calibration model provided by the International GNSS Service (IGS).

All files used are in : */01_DGNSS_positionning / 1_Input_files / 3_Auxiliary_files*

2. Options : validation method and uncertainty

Options & Settings

We needed to find the right settings and corrections to apply to obtain **consistent** and **accurate results**. We tried to retrieve the known coordinates of the base stations using their observation files.

The options used for the validation methods are in files : */01_DGNSS_positionning / 2_Methodology_options_RTKLIB / option_DD_TN01, option_DD_TN02, option_DD_IZAN* (with coordinates of base station in memory).

The screenshot shows the RTKLIB Options dialog box with the 'Setting1' tab selected. The 'Positioning Mode' is set to 'Static'. 'Frequencies' are set to 'L1+L2'. 'Filter Type' is 'Combined - no phase reset'. 'Elevation Mask' is '15'. 'Rec Dynamics / Earth Tides Correction' is 'ON'. 'Ionosphere Correction' is 'Estimate STEC'. 'Troposphere Correction' is 'Estimate ZTD+Grad'. 'Satellite Ephemeris/Clock' is 'Precise'. Under 'Excluded Satellites (+PRN: Included)', 'GPS', 'GLONASS', 'Galileo', 'QZSS', 'BDS', 'NavIC', and 'SBAS' are all checked. The 'Setting2' tab is also visible, showing 'Integer Ambiguity Res' set to '3', 'Ratio to Fix Ambiguity' set to '3', 'GLO HW Bias' set to '0', 'Min Lock / Elevation' set to '0', 'Min Fix / Elevation' set to '20', 'Slip Threshs' set to '0.000', 'Max Age of Diff' set to '30.0', 'Outlier Threshold' set to '30.0', '# of Filter Iter / Sync Solution' set to '1', 'Baseline Length Constraint' set to '0.000', 'Min Fix Sats / Min Hold Sats' set to '5', 'Min Drop Sats' set to '10', 'Max Pos Var for AR / AR Filter' set to '0.1000', and 'Hold Amb Var / Hold Amb Gain' set to '0.1000'.

The **important Options** to not forget are :

- Position mode : **Static**
- Satellite Ephemeris/Clock : **Precise** (to take into account the SP3 and CLK aux files)
- **Ionosphere** and **troposphere** correction : Estimated STEC & ZTD+Grad
- In **position** option : make sure it is the right **coordinates** for the base stations & click on **Antenna Type and select auto *** : it will read the head of the files and directly apply the antenna difference.
- In **Output** option : select solution for static mode : **All** (otherwise "single" it takes only the last line which is not always the best)

- In **files** : include the ERP file (/in EOP Data file) of the survey week and the antenna model file.

Validation of methodology

Station / Baseline	IZAN (from TN02)	IZAN (from TN01)	TN01 (from TN02)	TN01 (from IZAN)	TN02 (from IZAN)	TN02 (from TN01)
REAL altitude	2417,40444	2417,40444	51,75309	51,75309	54,41445	54,41445
D93	2417.3406	2417.3040	51.7938	51.8535	54.4781	54.3743
D94		2417.3059	51.7901	51.8516		
D129	2417.3314	2417.3064	51.7818	51.8511	54.4875	54.3845
D213	2417.3301				54.4889	
Difference / error	- 6.5 cm	- 10 cm	+ 3.5 cm	+ 10 cm	+ 6.5 cm	- 3.5 cm

By calculating the position (height) of each station, with a different base station and at 3 different dates, the results are consistent between each day (1 or 2 cm difference). The results are also consistent between each baseline : the difference is equal, with opposite signs.

(For the selection of the final value : I looked in the pos file and went to the end (=end of convergence) and selected not the last line but one of the 50 last with a ratio > 100 and stable with previous values.)

3. Survey

Now that we validated the consistency and accuracy of the methodology, we can proceed to the calculation of the stations. We will take into account that the accuracy will certainly be less because the observation files from the rover are only between 5 to 30 min recording files instead of 24h (= Rapid static). The recording time may influences :

- The convergence of the model to a stable value : check if solution is fixed or float
- The tropospheric effect estimations
- Dependent on Satellite geometry

This leads to observations being more sensible to local conditions of the moment and less stability.

Stations and dates of GPS measurements

Survey - Stations	Date	GPS date	Base station*	Note
3RDB	02/04/24	D93 / W2308 D2	IZAN	
CHIP	02/04/24	D93 / W2308 D2	IZAN	
LM (001-006)	02/04/2024	D93 / W2308 D2	IZAN	No LM04
IPNA	03/04/24	D94 / W2308 D3	TN01	
Fasnia (FA 00-07)	03/05/24	D124 / W2312 D5	IZAN	
Guimar (GUI 00-06)	06/05/24	D127 / W2313 D1	IZAN	IZAN far but better results
Guimar (GUI 07-19)	08/05/24	D129 / W2313 D3	TN01	**
Fasnia (FA 08-15)	09/05/24	D130 / W2313 D4	IZAN	
Mich (MI 00-09)	10/05/24	D131 / W2313 D5	TN02	MI08=09
Guimar (GUI 20-26)	03/06/24	D155 / W2317 D1	IZAN	
LM_G (07-12)	08/07/24	D190 / W2322 D1	IZAN	***
LM_G (13-19)	09/07/24	D191 / W2322 D2	IZAN	***
La Orotava (CARA + CAMI 00-09)	31/07/24	D213 / W2325 D3	IZAN	
CARA + GUI 30-34)	07/10/24	D281 / W2335 D1	IZAN	**
ORO 00-08	04/04/25	D94 / W2360 D5	TN02	

*The base station was chosen given its proximity to the observed station (in altitude and distance), using the same base for stations on the same day to keep coherence.

** Some day for GUI, no tripod that day for recording GPS so higher uncertainty.

*** LM_G : Some confusions between obs file and field notebook (corrected in names) and antenna height changed in obs file because the tripod was in soil.

GUI16 was deleted due to too high stu and no convergence result.

LM006 was in two times and the second was saved bc closer to the real point.

Results

After running the RTKPOST software for all the stations, we obtain a .pos file for every station with the results of coordinate solutions every second (static processing file) with uncertainties on the model (in [/3_Results/Raw_results](#)). We need to have just one single coordinate for each station.

For this, a python script was created ([/final_script_DGPS.ipynb](#)) to visualize the data : graphs of the calculated height of the stations as a function of time, as well as information (recording time, date, base station used), statistics (weighted mean, max/ min, standard deviation, standard error) and if the result of the model is fixed (green) or float (orange).

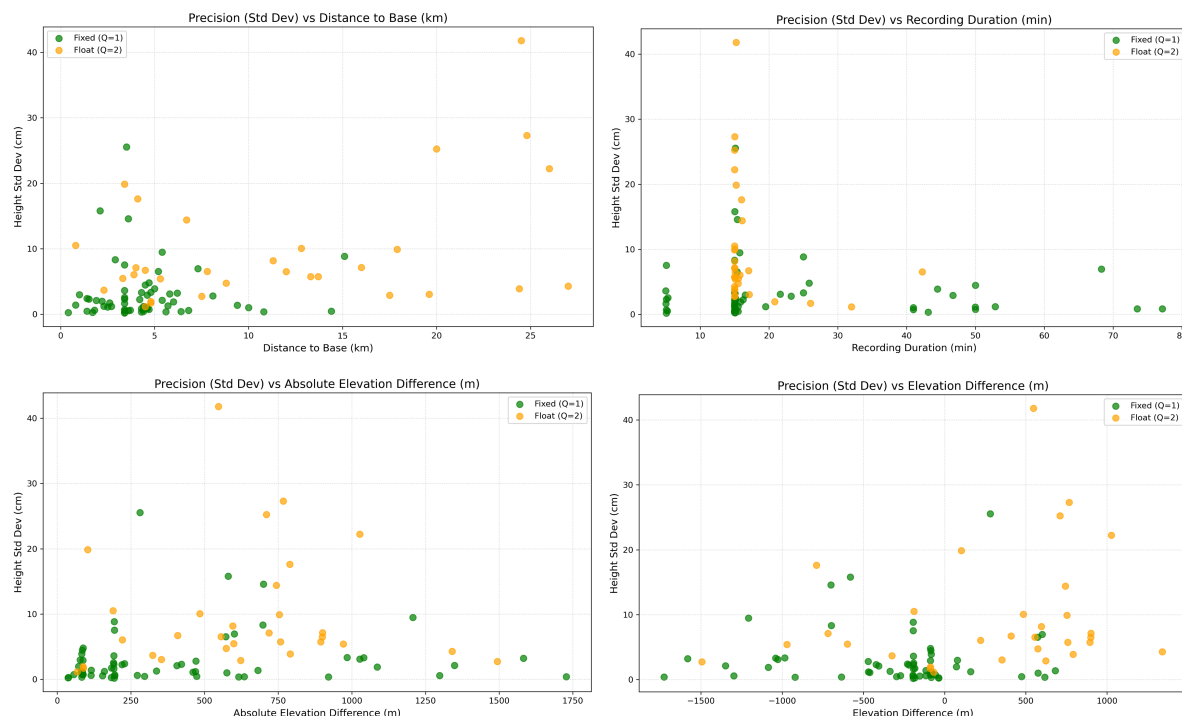
Finally, the output file is created ([/FINAL_RESULTS_GNSS_coordinates_stations.xlsx & .csv](#)) with, for each station, the date, the base station used, the recording time, the number of satellites, the weighted mean coordinates (latitude, longitude & heights) with their standard deviation, standard error and difference between mean and median (to validate the data), and also the distance between station and base station and their altitude difference, and the Q.

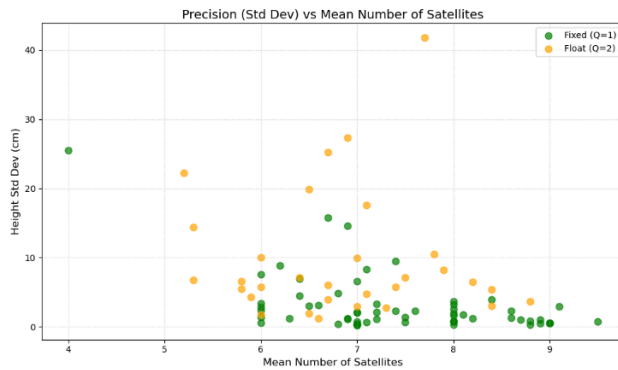
Uncertainties

The heights calculated by the software show a median standard deviation of 2.77 cm, which will be less than 0.01 mGal uncertainty.

min	0,2
25%	1,06
Mediana : 50%	2,77
75%	6,52
max	41,8
mean cm	5,2

The errors and uncertainty calculations were done, in order to correlate the error with a specific factor. The plots of the height error vs altitude & distance between stations, recording time and number of satellites does not show a very clear correlation.





- Standard deviation : precision and stability of the results
- Standard error : uncertainty of the statistics
- Difference mean-median : detect spikes and outliers values
- Q : value of confidence of statistical convergence. If Q=1 it means that the ratio > 3 = good confidence.

Documentation & links:

- Files:

IGN daily Base Station files : [Index of /ERGNSS/diario_30s](#) ;

Auxiliary Global files, Nasa : [CDDIS || archive | gnss |](#)

Antenna file : [Index of /pub/station/general](#) : [igs20.atx](#)

- Infos :

GPS Calendar : [GPS Calendar - National Geodetic Survey](#)

Coordinates : [MAGNET + Global GPS Network Map](#)

- Conversions :

Conversion .crx to .rnix : [CRX2RNX](#)

- Software :

RTKLIB 2.5.0 : [Release RTKLIB-EX 2.5.0 · rtklibexplorer/RTKLIB · GitHub](#)

Manual : [RTKLIB - Demo5](#)

Tuto and Figures : [gpspp.sakura.ne.jp/paper2005/GPS_RTKLIB_Seminor_1.pdf](#) (DGPS p67) & [gpspp.sakura.ne.jp/paper2005/GPS_RTKLIB_Seminor_2.pdf](#) (DD p8)

Tuto and demonstration : [www.rtklib.com/rtklib_tutorial.htm](#)

- Results and Files :

In / 01_DGNSS_positionning